Lesson Planning

Complete Lesson Plan Book

Class: Class 11 – FSc Pre-Engineering

Subject: Physics

Medium: English

Academic Year 2026-27

DM

This book is prepared as a teaching guide for teachers.

Table of Contents

Unit 1: Measurements

Unit 2: Vectors and Equilibrium

Unit 3: Motion and Force

Unit 4: Work and Energy

Unit 5: Circular Motion

Unit 6: Fluid Dynamics

Unit 7: Oscillations

Unit 8: Waves

Unit 9: Physical Optics

Unit 10: Optical Instruments

DM

Unit 1: Measurements

Learning Objectives

• Define and explain physical quantities in the context of Measurements.

• Differentiate between SI units and significant figures with suitable examples.

• Solve numerical problems related to precision and accuracy accurately.

• Apply the ideas of Measurements to everyday situations involving errors.

Key Concepts

• Meaning, definition and scope of physical quantities.

• Important properties, types and examples of SI units.

• Relationship between significant figures and precision and accuracy.

• Practical applications of Measurements, especially errors.

• Key terms and a summary of the main points of Measurements.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Measurements with a daily-life example.
2. Explain physical quantities on the board with definitions and clear diagrams.
3. Discuss SI units and significant figures using simple, relatable examples.
4. Involve the students through short questions on precision and accuracy.
5. Solve a numerical on errors together with the class.
6. Summarize the lesson and highlight the key points of Measurements.

Classroom Activities

• Group discussion: students compare SI units with significant figures.

• Board activity: solve a question on precision and accuracy in pairs.

• Quick quiz on the key terms of Measurements.

Assessment Questions

1. Define physical quantities.
2. Differentiate between SI units and significant figures.
3. Solve a numerical based on precision and accuracy.
4. Write a short note on errors.

Homework

• Read Measurements from the book and solve the exercise questions.

• Prepare short notes on physical quantities and SI units.

Time Allocation

4–6 periods of 40 minutes each

Unit 2: Vectors and Equilibrium

Learning Objectives

• Define and explain vector addition in the context of Vectors and Equilibrium.

• Differentiate between scalar product and vector product with suitable examples.

• Solve numerical problems related to torque accurately.

• Apply the ideas of Vectors and Equilibrium to everyday situations involving equilibrium conditions.

Key Concepts

• Meaning, definition and scope of vector addition.

• Important properties, types and examples of scalar product.

• Relationship between vector product and torque.

• Practical applications of Vectors and Equilibrium, especially equilibrium conditions.

• Key terms and a summary of the main points of Vectors and Equilibrium.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Vectors and Equilibrium with a daily-life example.
2. Explain vector addition on the board with definitions and clear diagrams.
3. Discuss scalar product and vector product using simple, relatable examples.
4. Involve the students through short questions on torque.
5. Solve a numerical on equilibrium conditions together with the class.
6. Summarize the lesson and highlight the key points of Vectors and Equilibrium.

Classroom Activities

• Group discussion: students compare scalar product with vector product.

• Board activity: solve a question on torque in pairs.

• Quick quiz on the key terms of Vectors and Equilibrium.

Assessment Questions

1. Define vector addition.
2. Differentiate between scalar product and vector product.
3. Solve a numerical based on torque.
4. Write a short note on equilibrium conditions.

Homework

• Read Vectors and Equilibrium from the book and solve the exercise questions.

• Prepare short notes on vector addition and scalar product.

Time Allocation

4–6 periods of 40 minutes each

Unit 3: Motion and Force

Learning Objectives

• Define and explain Laws of motion in the context of Motion and Force.
• Differentiate between momentum and friction with suitable examples.
• Solve numerical problems related to projectile motion accurately.
• Apply the ideas of Motion and Force to everyday situations involving impulse.

Key Concepts

• Meaning, definition and scope of Laws of motion.
• Important properties, types and examples of momentum.
• Relationship between friction and projectile motion.
• Practical applications of Motion and Force, especially impulse.
• Key terms and a summary of the main points of Motion and Force.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Motion and Force with a daily-life example.
2. Explain Laws of motion on the board with definitions and clear diagrams.
3. Discuss momentum and friction using simple, relatable examples.
4. Involve the students through short questions on projectile motion.
5. Solve a numerical on impulse together with the class.
6. Summarize the lesson and highlight the key points of Motion and Force.

Classroom Activities

• Group discussion: students compare momentum with friction.
• Board activity: solve a question on projectile motion in pairs.
• Quick quiz on the key terms of Motion and Force.

Assessment Questions

1. Define Laws of motion.
2. Differentiate between momentum and friction.
3. Solve a numerical based on projectile motion.
4. Write a short note on impulse.

Homework

• Read Motion and Force from the book and solve the exercise questions.
• Prepare short notes on Laws of motion and momentum.

Time Allocation

4–6 periods of 40 minutes each

Unit 4: Work and Energy

Learning Objectives

• Define and explain work done in the context of Work and Energy.

• Differentiate between kinetic energy and potential energy with suitable examples.

• Solve numerical problems related to power accurately.

• Apply the ideas of Work and Energy to everyday situations involving conservation of energy.

Key Concepts

• Meaning, definition and scope of work done.

• Important properties, types and examples of kinetic energy.

• Relationship between potential energy and power.

• Practical applications of Work and Energy, especially conservation of energy.

• Key terms and a summary of the main points of Work and Energy.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Work and Energy with a daily-life example.
2. Explain work done on the board with definitions and clear diagrams.
3. Discuss kinetic energy and potential energy using simple, relatable examples.
4. Involve the students through short questions on power.
5. Solve a numerical on conservation of energy together with the class.
6. Summarize the lesson and highlight the key points of Work and Energy.

Classroom Activities

• Group discussion: students compare kinetic energy with potential energy.

• Board activity: solve a question on power in pairs.

• Quick quiz on the key terms of Work and Energy.

Assessment Questions

1. Define work done.
2. Differentiate between kinetic energy and potential energy.
3. Solve a numerical based on power.
4. Write a short note on conservation of energy.

Homework

• Read Work and Energy from the book and solve the exercise questions.

• Prepare short notes on work done and kinetic energy.

Time Allocation

4–6 periods of 40 minutes each

Unit 5: Circular Motion

Learning Objectives

• Define and explain angular velocity in the context of Circular Motion.
• Differentiate between centripetal force and banking of roads with suitable examples.
• Solve numerical problems related to moment of inertia accurately.
• Apply the ideas of Circular Motion to everyday situations involving angular momentum.

Key Concepts

• Meaning, definition and scope of angular velocity.
• Important properties, types and examples of centripetal force.
• Relationship between banking of roads and moment of inertia.
• Practical applications of Circular Motion, especially angular momentum.
• Key terms and a summary of the main points of Circular Motion.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Circular Motion with a daily-life example.
2. Explain angular velocity on the board with definitions and clear diagrams.
3. Discuss centripetal force and banking of roads using simple, relatable examples.
4. Involve the students through short questions on moment of inertia.
5. Solve a numerical on angular momentum together with the class.
6. Summarize the lesson and highlight the key points of Circular Motion.

Classroom Activities

• Group discussion: students compare centripetal force with banking of roads.
• Board activity: solve a question on moment of inertia in pairs.
• Quick quiz on the key terms of Circular Motion.

Assessment Questions

1. Define angular velocity.
2. Differentiate between centripetal force and banking of roads.
3. Solve a numerical based on moment of inertia.
4. Write a short note on angular momentum.

Homework

• Read Circular Motion from the book and solve the exercise questions.
• Prepare short notes on angular velocity and centripetal force.

Time Allocation

4–6 periods of 40 minutes each

Unit 6: Fluid Dynamics

Learning Objectives

• Define and explain equation of continuity in the context of Fluid Dynamics.

• Differentiate between Bernoulli's equation and viscosity with suitable examples.

• Solve numerical problems related to Stokes' law accurately.

• Apply the ideas of Fluid Dynamics to everyday situations involving terminal velocity.

Key Concepts

• Meaning, definition and scope of equation of continuity.

• Important properties, types and examples of Bernoulli's equation.

• Relationship between viscosity and Stokes' law.

• Practical applications of Fluid Dynamics, especially terminal velocity.

• Key terms and a summary of the main points of Fluid Dynamics.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Fluid Dynamics with a daily-life example.
2. Explain equation of continuity on the board with definitions and clear diagrams.
3. Discuss Bernoulli's equation and viscosity using simple, relatable examples.
4. Involve the students through short questions on Stokes' law.
5. Solve a numerical on terminal velocity together with the class.
6. Summarize the lesson and highlight the key points of Fluid Dynamics.

Classroom Activities

• Group discussion: students compare Bernoulli's equation with viscosity.

• Board activity: solve a question on Stokes' law in pairs.

• Quick quiz on the key terms of Fluid Dynamics.

Assessment Questions

1. Define equation of continuity.
2. Differentiate between Bernoulli's equation and viscosity.
3. Solve a numerical based on Stokes' law.
4. Write a short note on terminal velocity.

Homework

• Read Fluid Dynamics from the book and solve the exercise questions.

• Prepare short notes on equation of continuity and Bernoulli's equation.

Time Allocation

4–6 periods of 40 minutes each

Unit 7: Oscillations

Learning Objectives

• Define and explain simple harmonic motion in the context of Oscillations.
• Differentiate between mass-spring system and simple pendulum with suitable examples.
• Solve numerical problems related to energy in SHM accurately.
• Apply the ideas of Oscillations to everyday situations involving damped oscillations.

Key Concepts

• Meaning, definition and scope of simple harmonic motion.
• Important properties, types and examples of mass-spring system.
• Relationship between simple pendulum and energy in SHM.
• Practical applications of Oscillations, especially damped oscillations.
• Key terms and a summary of the main points of Oscillations.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Oscillations with a daily-life example.
2. Explain simple harmonic motion on the board with definitions and clear diagrams.
3. Discuss mass-spring system and simple pendulum using simple, relatable examples.
4. Involve the students through short questions on energy in SHM.
5. Solve a numerical on damped oscillations together with the class.
6. Summarize the lesson and highlight the key points of Oscillations.

Classroom Activities

• Group discussion: students compare mass-spring system with simple pendulum.
• Board activity: solve a question on energy in SHM in pairs.
• Quick quiz on the key terms of Oscillations.

Assessment Questions

1. Define simple harmonic motion.
2. Differentiate between mass-spring system and simple pendulum.
3. Solve a numerical based on energy in SHM.
4. Write a short note on damped oscillations.

Homework

• Read Oscillations from the book and solve the exercise questions.
• Prepare short notes on simple harmonic motion and mass-spring system.

Time Allocation

4–6 periods of 40 minutes each

Unit 8: Waves

Learning Objectives

• Define and explain wave types in the context of Waves.
• Differentiate between speed of sound and Doppler effect with suitable examples.
• Solve numerical problems related to stationary waves accurately.
• Apply the ideas of Waves to everyday situations involving resonance.

Key Concepts

• Meaning, definition and scope of wave types.
• Important properties, types and examples of speed of sound.
• Relationship between Doppler effect and stationary waves.
• Practical applications of Waves, especially resonance.
• Key terms and a summary of the main points of Waves.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Waves with a daily-life example.
2. Explain wave types on the board with definitions and clear diagrams.
3. Discuss speed of sound and Doppler effect using simple, relatable examples.
4. Involve the students through short questions on stationary waves.
5. Solve a numerical on resonance together with the class.
6. Summarize the lesson and highlight the key points of Waves.

Classroom Activities

• Group discussion: students compare speed of sound with Doppler effect.
• Board activity: solve a question on stationary waves in pairs.
• Quick quiz on the key terms of Waves.

Assessment Questions

1. Define wave types.
2. Differentiate between speed of sound and Doppler effect.
3. Solve a numerical based on stationary waves.
4. Write a short note on resonance.

Homework

• Read Waves from the book and solve the exercise questions.
• Prepare short notes on wave types and speed of sound.

Time Allocation

4–6 periods of 40 minutes each

Unit 9: Physical Optics

Learning Objectives

• Define and explain interference in the context of Physical Optics.

• Differentiate between Young's experiment and diffraction with suitable examples.

• Solve numerical problems related to polarization accurately.

• Apply the ideas of Physical Optics to everyday situations involving thin film interference.

Key Concepts

• Meaning, definition and scope of interference.

• Important properties, types and examples of Young's experiment.

• Relationship between diffraction and polarization.

• Practical applications of Physical Optics, especially thin film interference.

• Key terms and a summary of the main points of Physical Optics.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Physical Optics with a daily-life example.
2. Explain interference on the board with definitions and clear diagrams.
3. Discuss Young's experiment and diffraction using simple, relatable examples.
4. Involve the students through short questions on polarization.
5. Solve a numerical on thin film interference together with the class.
6. Summarize the lesson and highlight the key points of Physical Optics.

Classroom Activities

• Group discussion: students compare Young's experiment with diffraction.

• Board activity: solve a question on polarization in pairs.

• Quick quiz on the key terms of Physical Optics.

Assessment Questions

1. Define interference.
2. Differentiate between Young's experiment and diffraction.
3. Solve a numerical based on polarization.
4. Write a short note on thin film interference.

Homework

• Read Physical Optics from the book and solve the exercise questions.

• Prepare short notes on interference and Young's experiment.

Time Allocation

4–6 periods of 40 minutes each

Unit 10: Optical Instruments

Learning Objectives

• Define and explain human eye in the context of Optical Instruments.
• Differentiate between microscope and telescope with suitable examples.
• Solve numerical problems related to magnifying glass accurately.
• Apply the ideas of Optical Instruments to everyday situations involving resolving power.

Key Concepts

• Meaning, definition and scope of human eye.
• Important properties, types and examples of microscope.
• Relationship between telescope and magnifying glass.
• Practical applications of Optical Instruments, especially resolving power.
• Key terms and a summary of the main points of Optical Instruments.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Optical Instruments with a daily-life example.
2. Explain human eye on the board with definitions and clear diagrams.
3. Discuss microscope and telescope using simple, relatable examples.
4. Involve the students through short questions on magnifying glass.
5. Solve a numerical on resolving power together with the class.
6. Summarize the lesson and highlight the key points of Optical Instruments.

Classroom Activities

• Group discussion: students compare microscope with telescope.
• Board activity: solve a question on magnifying glass in pairs.
• Quick quiz on the key terms of Optical Instruments.

Assessment Questions

1. Define human eye.
2. Differentiate between microscope and telescope.
3. Solve a numerical based on magnifying glass.
4. Write a short note on resolving power.

Homework

• Read Optical Instruments from the book and solve the exercise questions.
• Prepare short notes on human eye and microscope.

Time Allocation

4–6 periods of 40 minutes each